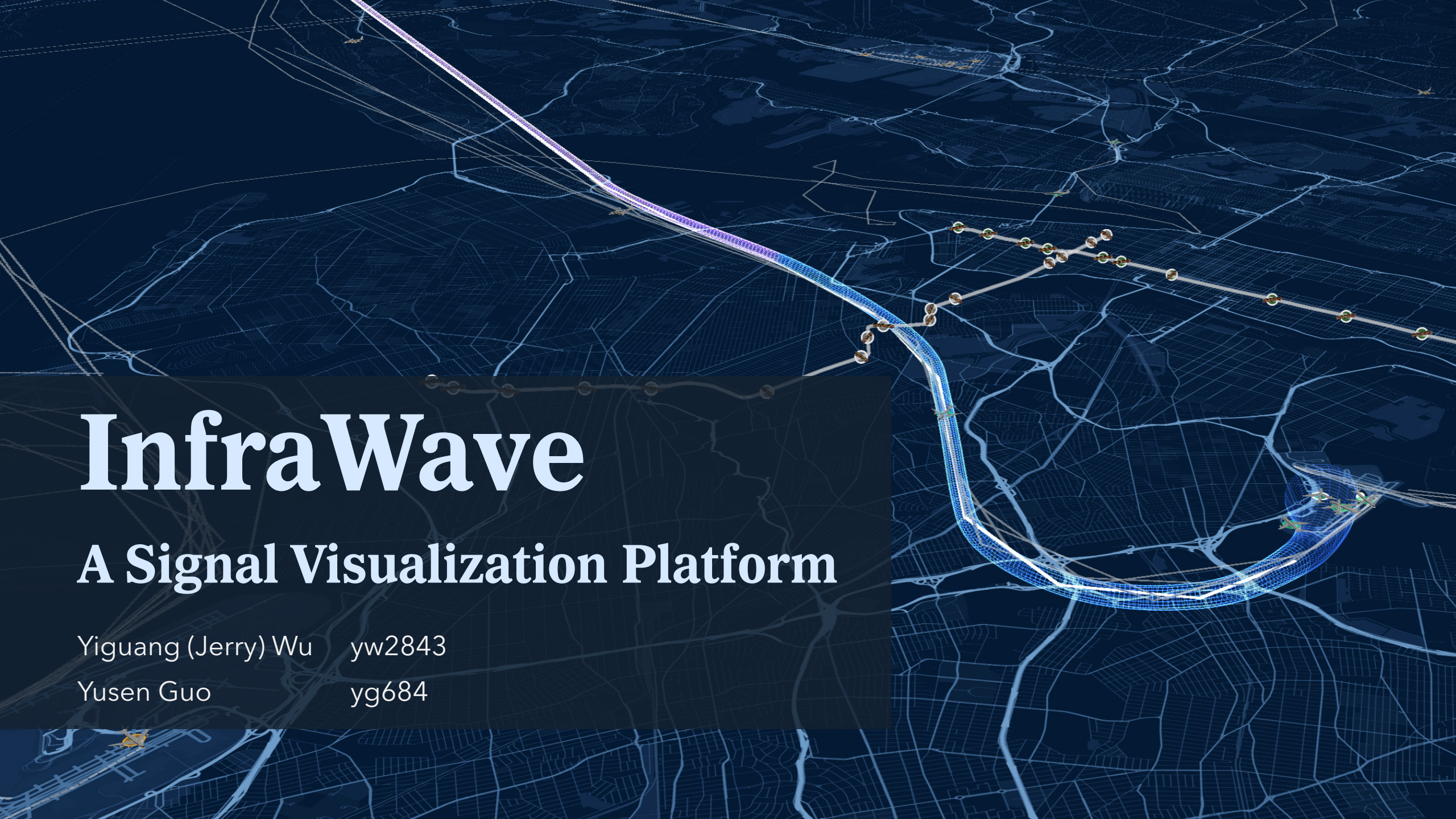




InfraWave

A Signal Visualization Platform

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What is it for?

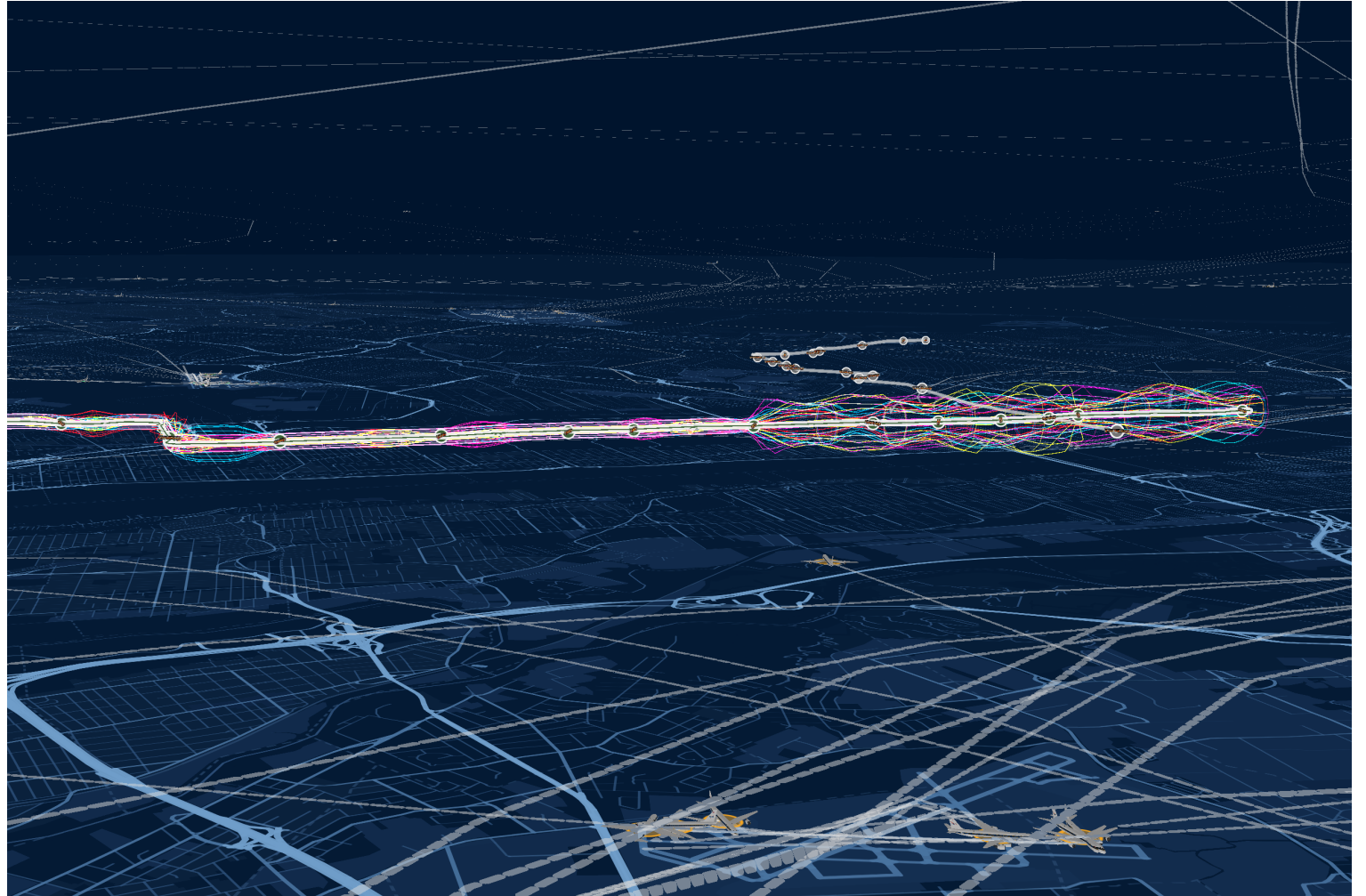
■ Signals along the trips

Concept & Inspiration

Telecommunication systems are essential to transportation, but their performance is rarely represented and difficult to perceive spatially.

InfraWave helps users visualize the **invisible signal infrastructures** behind their journeys.

It has information related to the **subway telecommunication signals** and **plane VHF radio signals**, and there are potentials to expand.



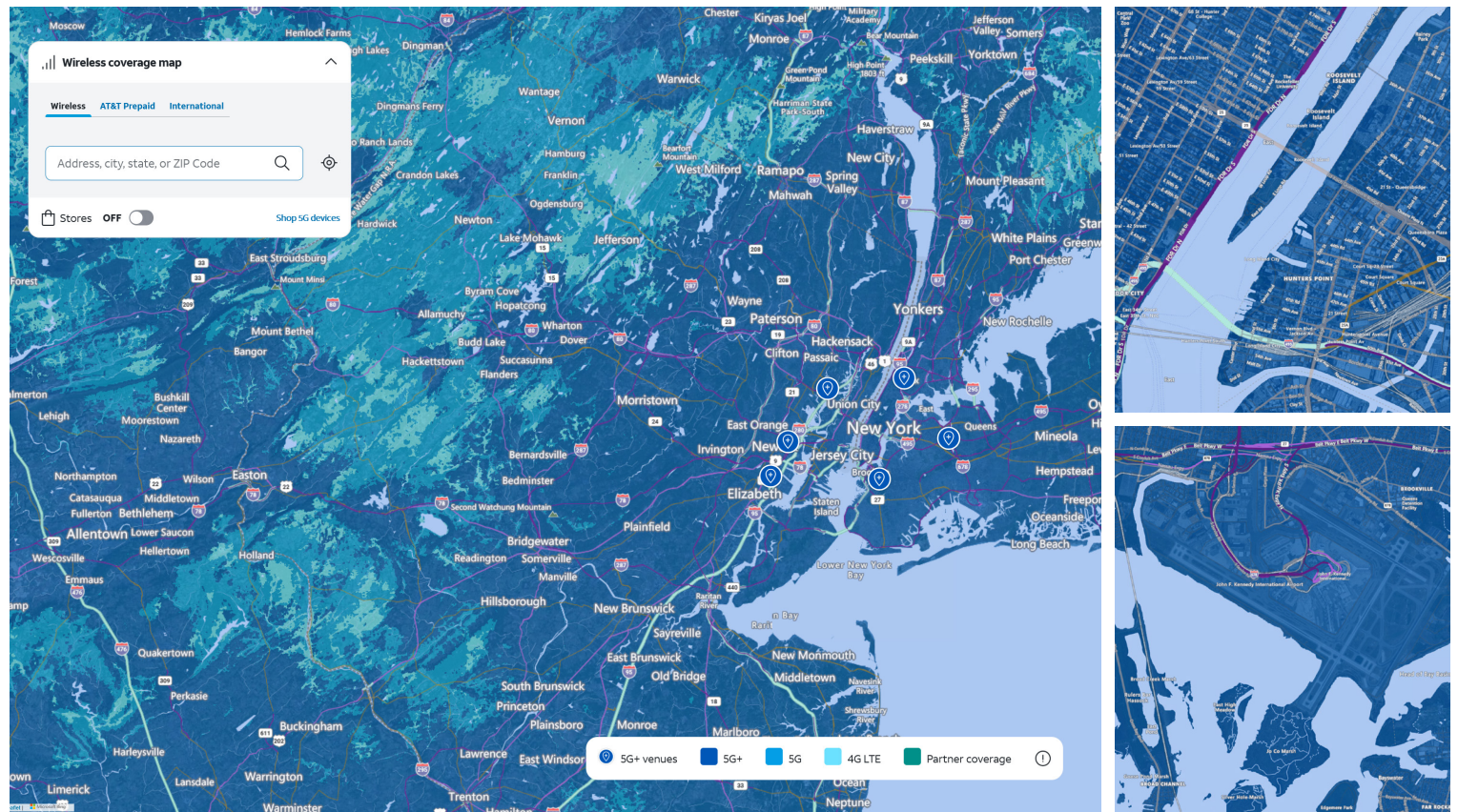
■ Measurement differs from installation

Site, Stakeholders, and Speculation

The project focuses on a missing dataset in the field of signal infrastructure: **The signal strength experienced inside moving transportation systems.**

Official coverage data is collected and modeled from a **statistic, planar, and car-oriented** point of view. It fails to account for people moving fast, being above / below ground, or occupying places cars can not reach.

L: AT&T coverage data & how their “covered” places are not covered.



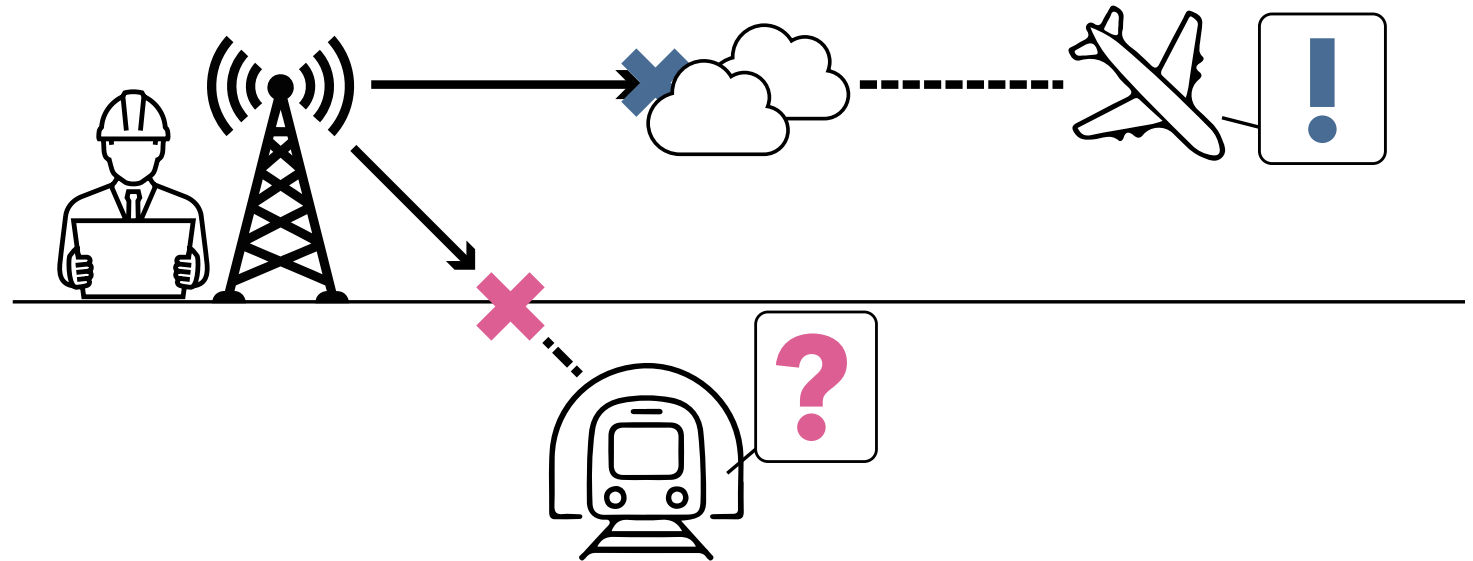
■ Subways and planes

Site, Stakeholders, and Speculation

InfraWave helps citizens and decision makers. This includes:

Transportation agencies;
Infrastructure operators;
Internet service providers;
Airline companies;
...

It reveals how signal infrastructure behaves across routes, locations, times, operators, and transportation systems.



How did we prototype?

■ Fetch open data

Data Collection

We collected data from:

OpenSky Network:	Real time Flight data
FAA:	Frequency matching regulation & Flight data regulations
MTA GTFS -Realtime Feed:	Real time subway locations
Subspotting:	Measured signal strength in NYC subways(2017).

Most datasets are well maintained. The only poorly maintained data is the signal strength we collected from Subspotting, which has not been updated since 2017.

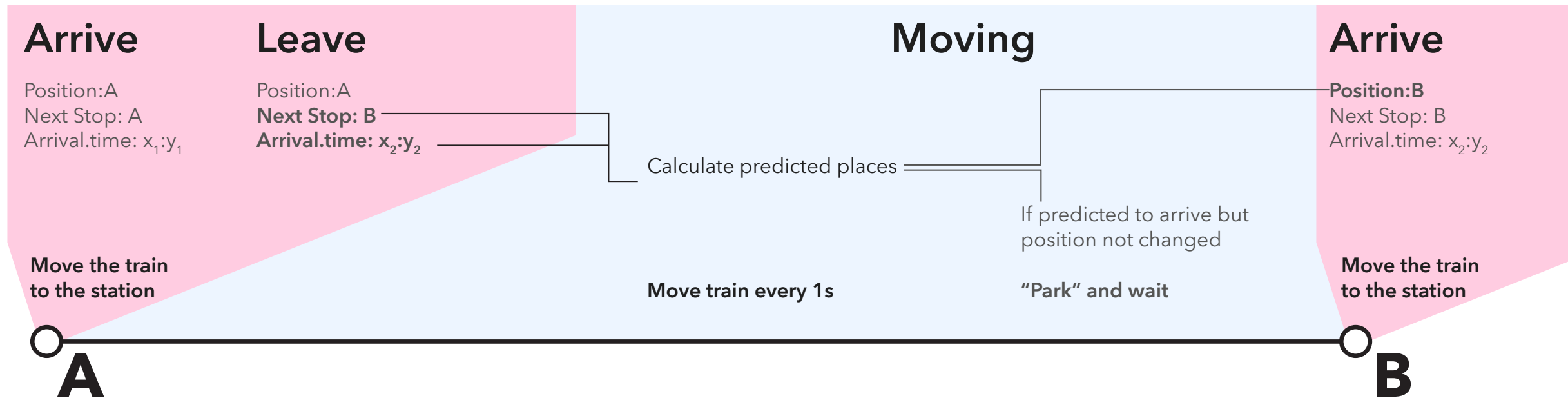
■ Estimating train positions

Processing / Analysis

■ Data

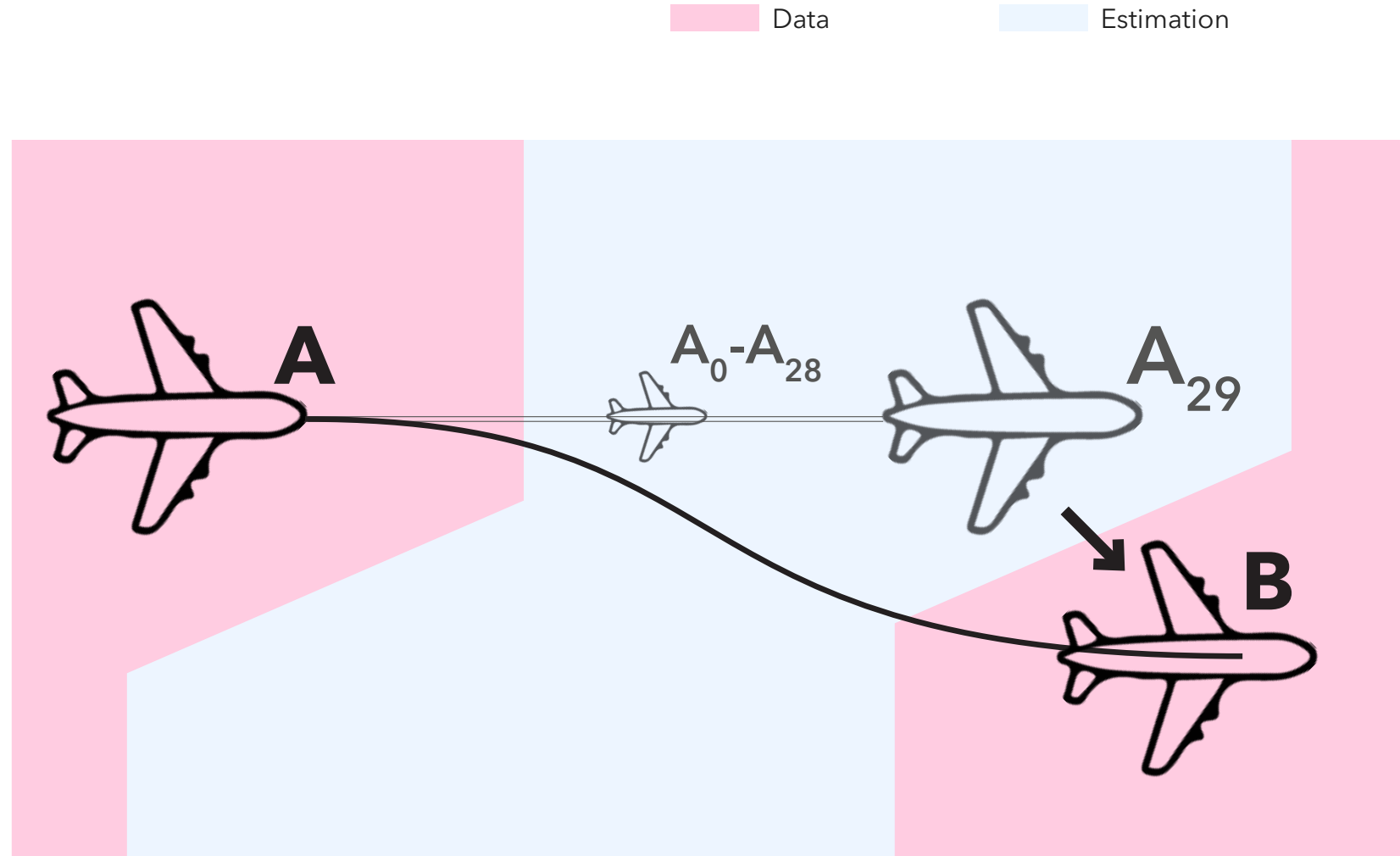
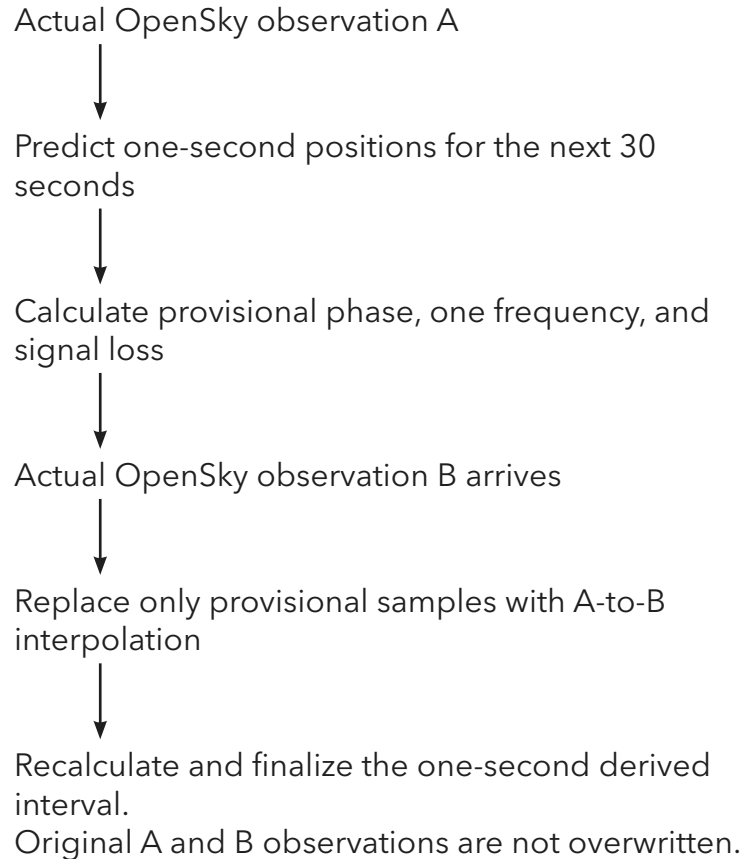
■ Estimation

The real time subway location is always snapped onto stations. We developed an algorithm to predict its position on the tracks based on the arrival time our MTA feed provided.



■ Real-time Flight Location

Processing / Analysis



■ Prototype Location Choice

Processing / Analysis

The flight layer focuses on VHF radio communication between aircraft and air traffic control facilities.

Flight stage	JFK — Kennedy	<u>LGA — LaGuardia</u>	EWR — Newark
At gate: Clearance Delivery	135.05	<u>135.2</u>	118.85
Taxiing: Ground Control	121.9	<u>121.7</u>	121.8; secondary 126.15
Take off / Landing: Tower	119.1 or 123.9	<u>118.7</u>	118.3; secondary 134.05
Approaching Airport	New York Approach 128.125	<u>New York Approach 120.8;</u> <u>some sectors 120.05</u>	Newark Approach 128.55, 132.8, 127.6, or 132.7, depending on sector
Immediately After Departure	New York Departure 135.9	<u>New York Departure 120.4</u>	Newark Departure 119.2
Leaving NYC Terminal Region	New York Center frequency assigned by ATC	New York Center frequency assigned by ATC	New York Center frequency assigned by ATC

■ Flight Frequency Strength Prediction

Processed Frequency Matching Rule in Prototype

28 Day NASR Subscription, Federal Aviation Administration.

Inferred phase	Conditions	Assigned frequency
Ground/taxi	Within 1.5 NM of LGA and on ground, or ≤ 250 ft and ≤ 50 kt	**121.7 MHz** LGA Ground
Final approach	Arrival, descending ≥ 200 fpm, within 8 NM, $\leq 3,000$ ft	**118.7 MHz** LGA Tower
Initial departure	Departure, climbing ≥ 200 fpm, within 5 NM, $\leq 3,000$ ft	**118.7 MHz** LGA Tower
Arrival approach	Arrival, descending ≥ 200 fpm, within 40 NM, $\leq 10,000$ ft	**120.8 MHz** N90 Approach
Departure climb	Departure, climbing ≥ 200 fpm, within 40 NM, $\leq 15,000$ ft	**120.4 MHz** N90 Departure

■ Flight Frequency Strength Prediction

Signal Loss Calculation - Relative Signal Strength

Calculate the signal loss for Free-space path loss and Building diffraction loss. The signal strength is not the real strength which is received by the flight, but a prediction strength.

With more accurate and entire dataset about the signal launching, the accurate signal strength should be able to be calculated and predicted.

1. It calculates the 3D slant distance between the aircraft and a synthetic LGA transmitter positioned 50 m above the airport.
2. It calculates free-space path loss:

$$FSPL = 32.45 + 20 \log_{10}(distance_{km}) + 20 \log_{10}(frequency_{MHz})$$

3. It checks buildings intersecting the signal path and estimates single knife-edge diffraction loss. Only the most dominant building is used.
4. Total modeled loss is:

$$Total Loss = FSPL + Building Loss$$

Lower total loss represents a stronger modeled signal path.

5. Relative signal strength is measured against the lowest-loss point:

$$Relative Signal_{dB} = Minimum Loss - Current Loss$$

$$Relative Power_{\%} = 100 \times 10^{Relative Signal_{dB}/10}$$

Therefore, 0 dB equals 100% of the reference power, -3 dB approximately 50%, -10 dB 10%, and -20 dB 1%.

These results represent **theoretical path loss and relative power**, not actual measured RSSI or received power in dBm.



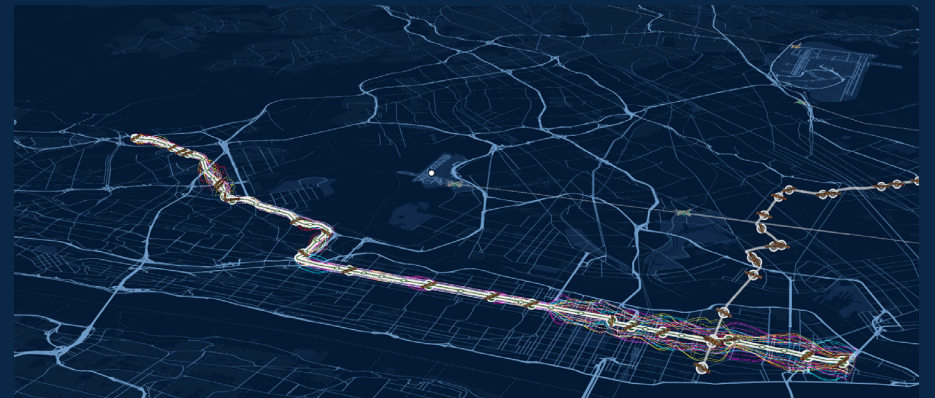
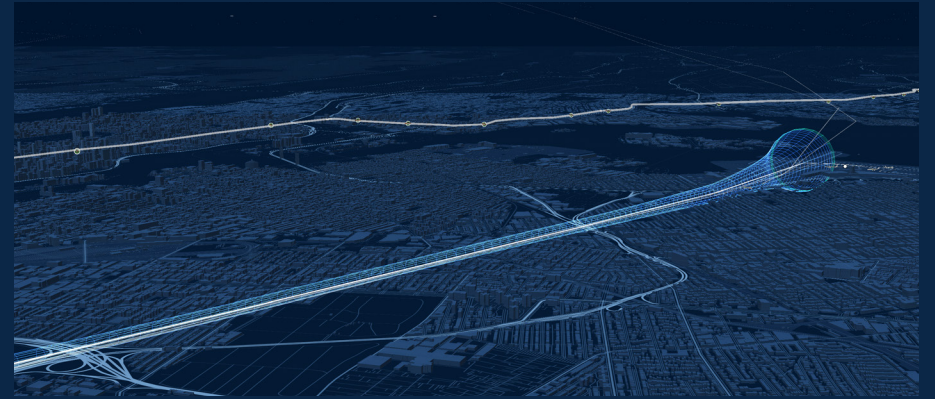
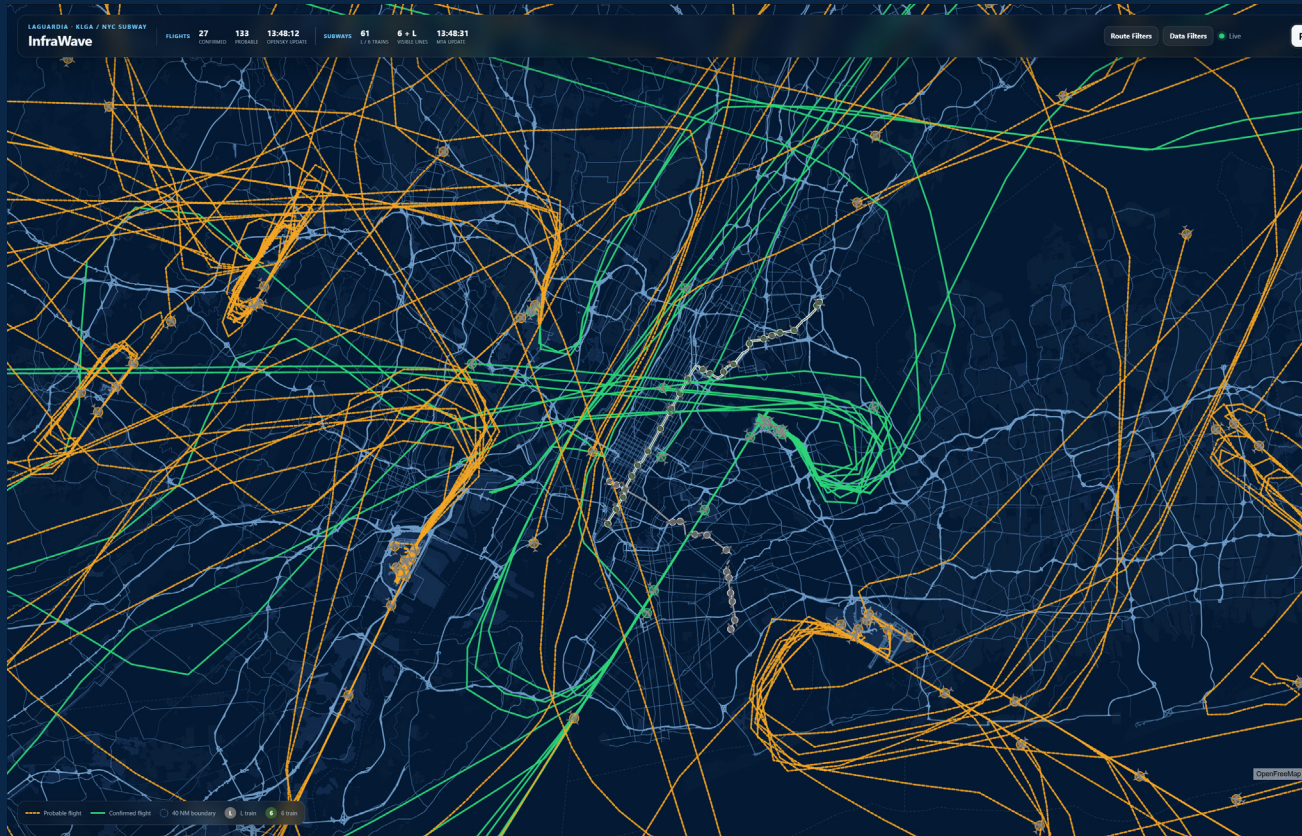
Now to the website...

Click here to experience yourself!

https://industry-engagement.github.io/Platform_Signal/

■ Select your planes & subways

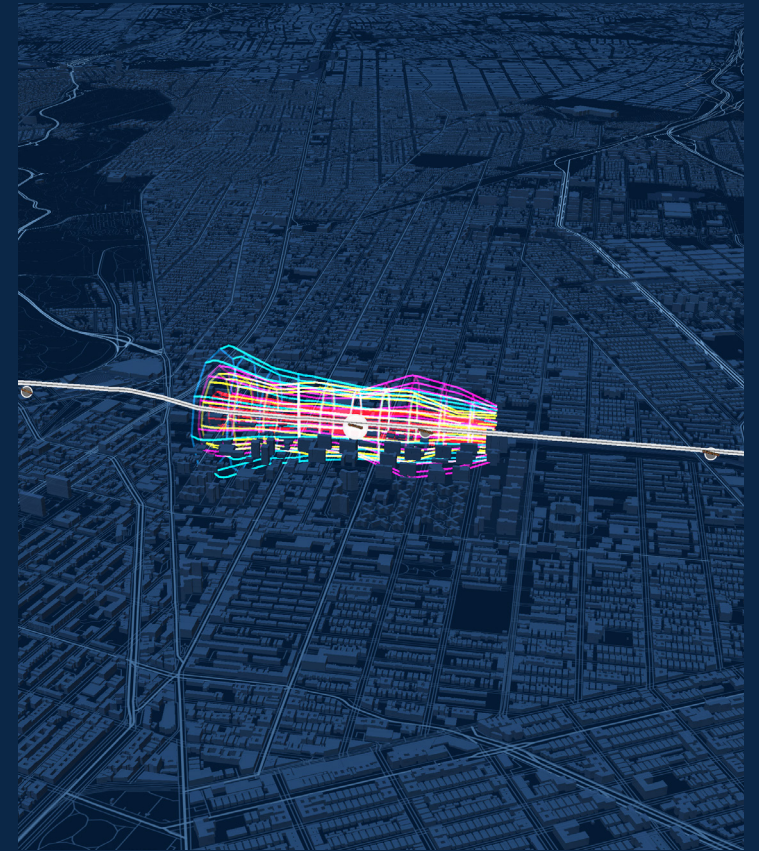
Visualization



■ “Signal Tubes”

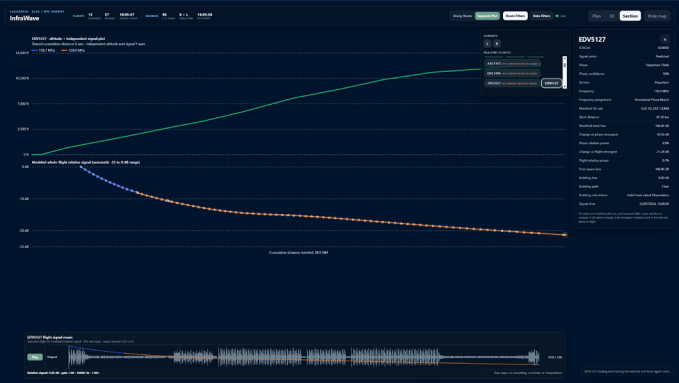
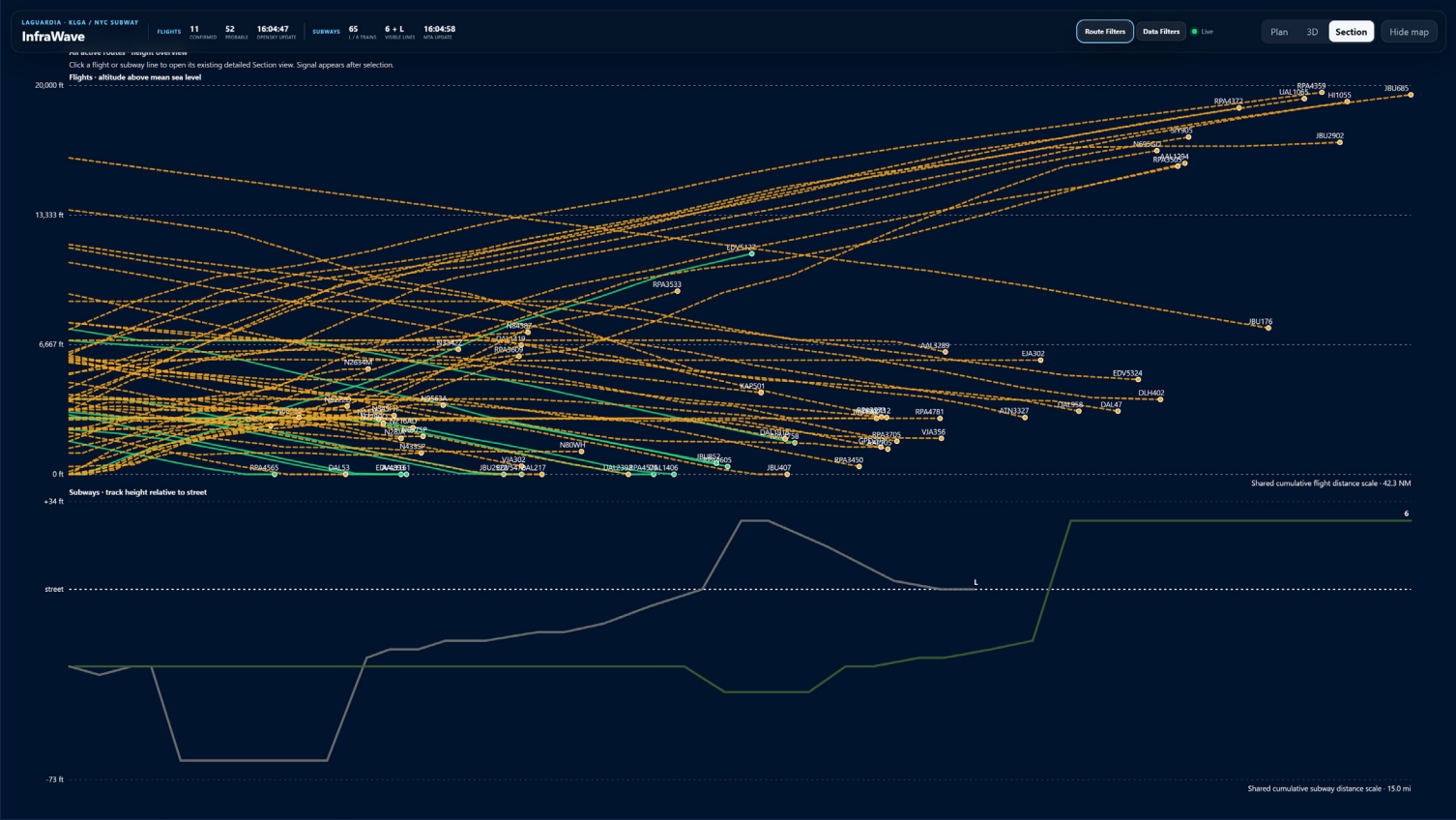
Visualization

To combine signal visualization with geographical information, we showed signals as glowing “tubes” in our website. The stronger the signal strength is, the larger its diameter would be. We also color-coded tubes by their frequencies / carriers.



Section view

Visualization



■ Experience signals acoustically

Visualization

We also tried using sounds to represent signal strength.

The volume of the music changes as signal strengthens or weakens at different places.



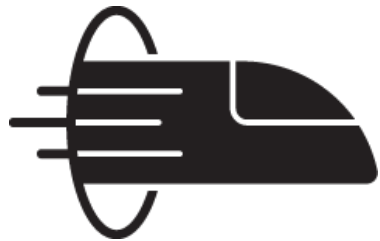
Long-term Management

■ Future

Collection / Processing / Analysis

A re-measurement of subway signal strength could be conducted to replace the outdated dataset for subway connections (2017).

Surveys could also be conducted to collect data from the public, providing a forecast for subways across the city.



Each participating flight could compare expected and actual signal performance and **improve the prediction model**.

Over time, the platform would construct a **3D dataset** of VHF communication performance across NYC airspace.



The **platform** will function as a **digital twin** of New York City transportation that specifies telecommunication. It will monitor changes, predict performances, and help agencies and operators determine where infrastructure should be upgraded.

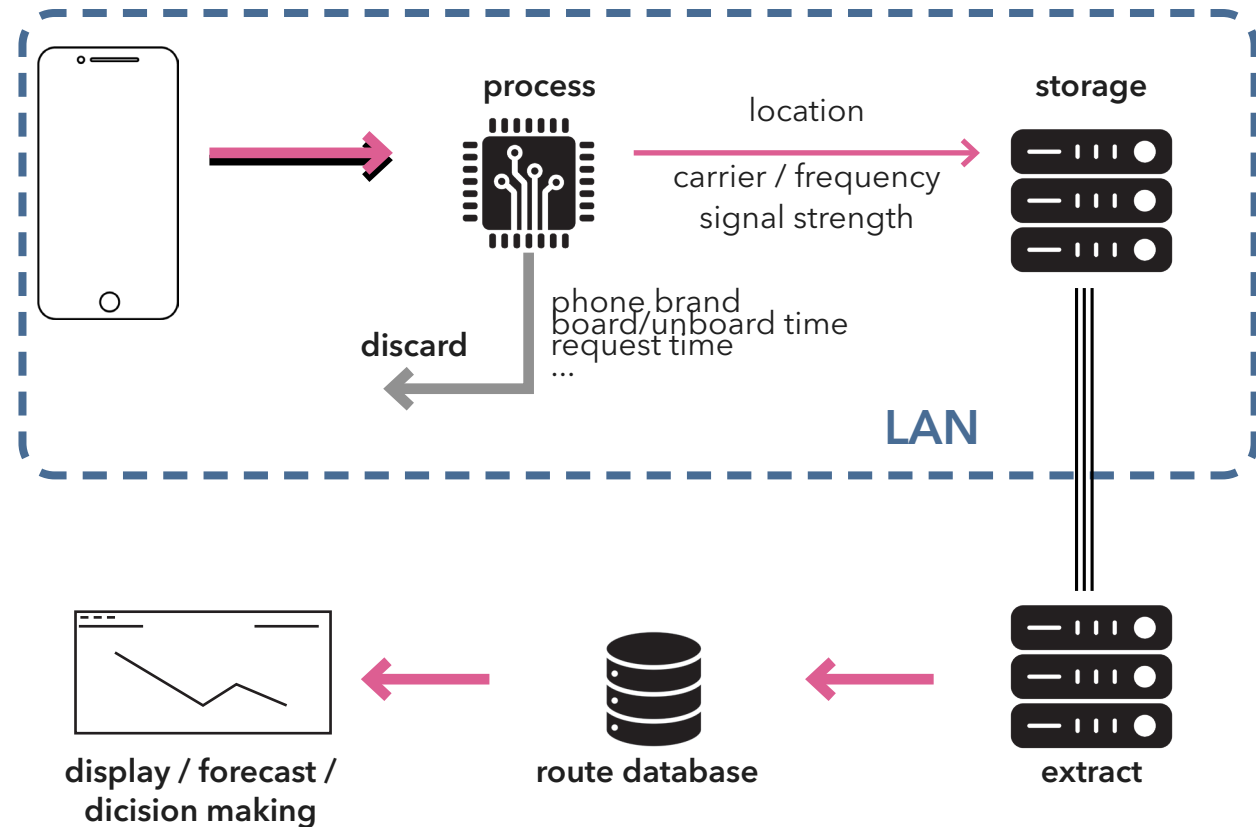


■ Where will they be housed?

Storage & Infrastructure

A Local Area Network (LAN) could be built on public transportation vehicles like subways and planes so that surveys could be conducted even without signal.

Fields needed only for processing are **discarded** immediately after use. Only location, carrier, and signal strength are encrypted and stored, so no identifying information is retained.



■ Who sees, who works, who tells your story

Governance

In the future, Rather than controlled by one company, the platform should be **maintained through a partnership** consists of:

MTA;
Airlines Companies;
Internet Providers;
...

Ethics

Different access levels should separate public information from sensitive operational data, to make sure the data and privacy security.

In order to make sure the privacy security and Signal data should be **anonymized, securely stored and only used for infrastructure improvement.**

Maintenance

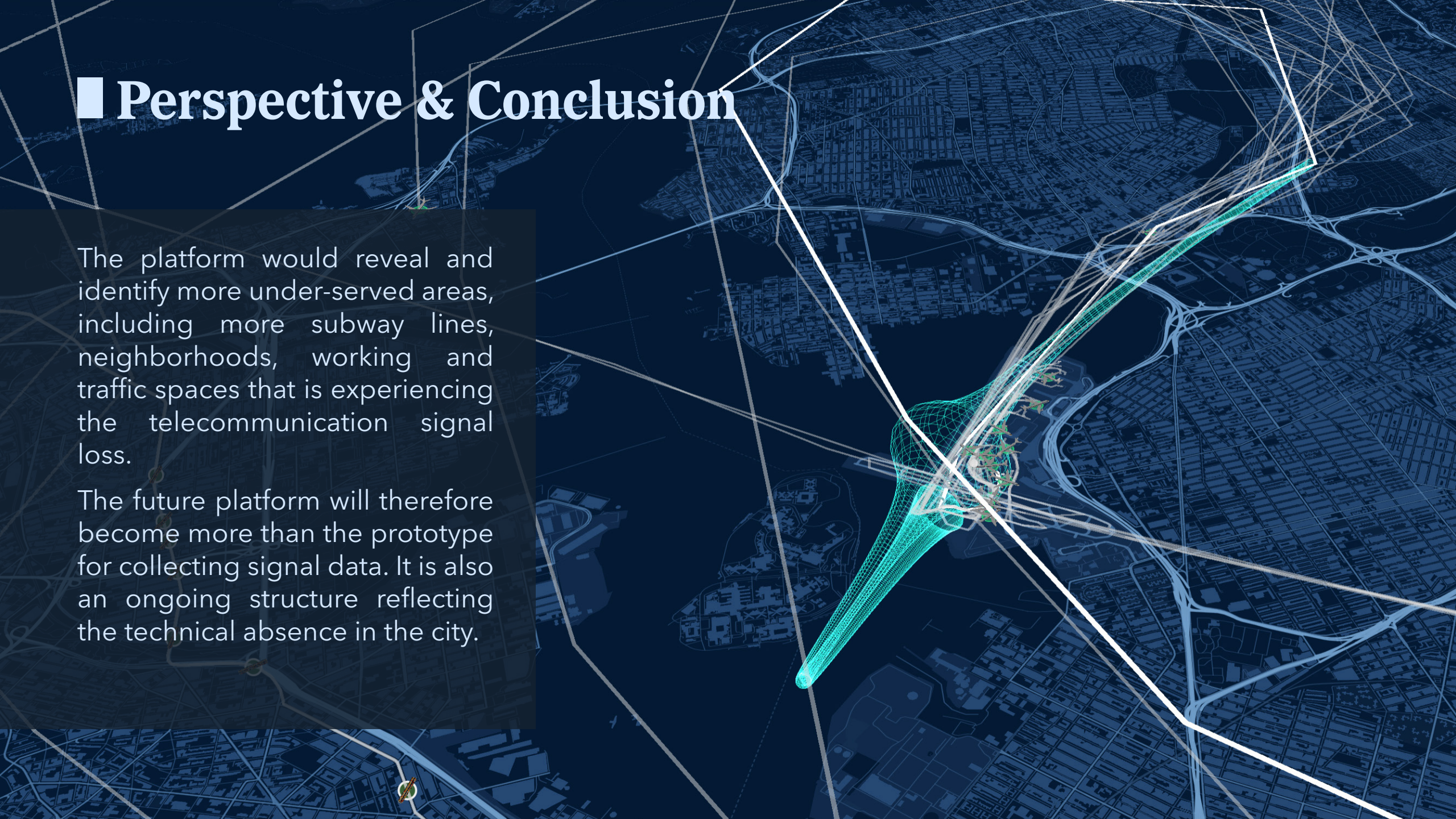
The platform requires **continuous care** through:

Sensor calibration,
Model validation,
Cybersecurity,
Transparent governance,
Equitable investment
Clear accountability.

■ Perspective & Conclusion

The platform would reveal and identify more under-served areas, including more subway lines, neighborhoods, working and traffic spaces that is experiencing the telecommunication signal loss.

The future platform will therefore become more than the prototype for collecting signal data. It is also an ongoing structure reflecting the technical absence in the city.



Thank You!

InfraWave - A Signal Visualization Platform

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